

Countable Spectral Extensions: One Positive Case and Empty-Interior Obstructions

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Source Question

This packet concerns Remark 4.21(iii) in Ryshard-Pavel Kostecki, *Vainberg–Bregman relative entropy and quasinonexpansive operators*, arXiv:2511.14873 [1].

homeomorphy of $\mathcal{X}$ follows from [27, Props. 3.3, 3.4, Lem. 3.5]. □

- Remark 4.21.** (i) The condition on $T : \mathcal{H} \rightarrow \mathcal{H}$ assumed in Proposition 4.15 holds, in particular, when T is positive semidefinite and invertible [290, Ex. 3.2], as well as when $\mathcal{H}$ is finite dimensional and T is symmetric and positive definite [291, p. 64] (this case goes back to [68, p. 15]). For $T = \mathbb{I}_{\mathcal{H}}$ with arbitrary dimensional $\mathcal{H}$, one obtains $D_{\Psi_T} = D_{\varphi_{1,1/2}}$ (see Remark 3.37.(x)), which was considered as an example (for $\dim \mathcal{H} < \infty$) already in [67, p. 1021] and [68, §2.1].
- (ii) The formula (193) has appeared earlier in [269, p. 376], however without using spectral convex functions (the Taylor expansion formula, and the finite dimensional quantum analogue $D_{\Psi}^{\text{tr}\mathcal{H}}$ of the Brunk–Ewing–Utz functional were used instead). Formula (195) is new.
- (iii) The main open problem posed by Proposition 4.20, is it provide examples of unitarily invariant Euler–Legendre functions $f \circ \tilde{\lambda}$ on the suitable separable reflexive spaces $(\mathfrak{G}(\mathcal{H}), \|\cdot\|_{\mathfrak{G}(\mathcal{H})})$. In particular, is it so for the countable extensions of functions listed in Proposition 4.17?

The source asks for examples of unitarily invariant Euler–Legendre functions $f \circ \tilde{\lambda}$ on suitable separable reflexive spaces of compact operators, and asks in particular whether this works for the countable extensions of the finite spectral functions listed in Proposition 4.17.

Claimed Status

Candidate partial result, likely valid pending human review. The packet classifies the direct countable extensions of the finite examples in Proposition 4.17 into the one full-domain positive case and the domain-restricted cases excluded by empty interior. The broader problem of finding genuinely new non- Ψ_{φ} examples remains open.

Proof Intuition

The finite list has two different behaviours. The full-domain power example is just the usual Legendre function $(1/p)\|x\|_p^p$ on ℓ_p and on the Schatten class S_p , so it survives in infinite dimension. The entropy examples whose domains are the positive cone, the positive definite cone, or the operator interval $0 \leq x \leq I$ do not survive the same way. In infinite-dimensional compact-operator

ideals, every positive compact operator has arbitrarily small spectral directions. A tiny negative rank-one perturbation in such a direction is arbitrarily small in any compact ideal norm, but it exits the positive cone. Thus those natural domains have empty norm interior, so they cannot meet the nonempty-interior hypotheses used in Proposition 4.20.

Setup

Let $\mathcal{H}$ be an infinite-dimensional separable Hilbert space. Let $\mathcal{I}(\mathcal{H})$ be any Banach ideal of compact operators on $\mathcal{H}$ whose norm is unitarily invariant and finite on rank-one operators; this includes the Schatten classes and the rearrangement-invariant compact ideals appearing in Proposition 4.20 of the source paper. Let

$$\mathcal{I}(\mathcal{H})^{\text{sa}} = \{x \in \mathcal{I}(\mathcal{H}) : x = x^*\}$$

with the ideal norm.

For a Borel scalar function ϕ on a real interval, the naive countable spectral extension of the finite-dimensional functional $\sum_i \phi(\lambda_i)$ has natural domain determined by the spectral constraint on the compact self-adjoint operator. For the domain-restricted examples in Proposition 4.17 and Example 2.5 of the source, these domains are contained in one of:

$$\mathcal{I}(\mathcal{H})^+, \quad \mathcal{I}(\mathcal{H})_0^+, \quad \{x \in \mathcal{I}(\mathcal{H})^{\text{sa}} : 0 \leq x \leq I\}.$$

Here $\mathcal{I}(\mathcal{H})^+$ denotes positive compact operators in the ideal, and $\mathcal{I}(\mathcal{H})_0^+$ denotes the strictly positive part of that cone when such a condition is imposed by the formula.

Lemma 1 (The full-domain power extension works). *Let $1 < p < \infty$ and define $f : \ell_p \rightarrow \mathbb{R}$ by*

$$f(x) = \frac{1}{p} \sum_{n=1}^{\infty} |x_n|^p = \frac{1}{p} \|x\|_p^p.$$

Then f is Euler–Legendre on ℓ_p . Moreover, for the Schatten class $S_p(\mathcal{H})^{\text{sa}}$ one has

$$(f \circ \tilde{\lambda})(a) = \frac{1}{p} \|a\|_{S_p}^p,$$

and $f \circ \tilde{\lambda}$ is a unitarily invariant Euler–Legendre function on $S_p(\mathcal{H})^{\text{sa}}$.

Proof. Since $1 < p < \infty$, both ℓ_p and its dual ℓ_q are reflexive, strictly convex, and smooth. The subdifferential of $f = (1/p)\|\cdot\|_p^p$ is the usual duality map

$$\nabla f(x) = (|x_n|^{p-2}x_n)_n \in \ell_q, \quad q = \frac{p}{p-1},$$

with the usual interpretation at $x_n = 0$. Its Fenchel dual is $f^*(y) = (1/q)\|y\|_q^q$, whose gradient is the inverse duality map. Thus f and f^* are Gateaux differentiable on their whole spaces and strictly convex; equivalently, f is Euler–Legendre.

For $a \in S_p(\mathcal{H})^{\text{sa}}$, $\tilde{\lambda}(a)$ is the signed decreasing rearrangement of the eigenvalues of a , counted with multiplicity, and therefore

$$\|\tilde{\lambda}(a)\|_{\ell_p} = \|a\|_{S_p}.$$

Hence $(f \circ \tilde{\lambda})(a) = (1/p)\|a\|_{S_p}^p$. The same argument as above, now applied to the uniformly convex and smooth Schatten class S_p , shows that this function is Euler–Legendre on $S_p(\mathcal{H})^{\text{sa}}$. Unitary invariance follows from invariance of the singular/eigenvalue list under unitary conjugacy. $\square$

Lemma 2. *The positive cone $\mathcal{I}(\mathcal{H})^+$ has empty interior in $\mathcal{I}(\mathcal{H})^{\text{sa}}$.*

Proof. Fix $a \in \mathcal{I}(\mathcal{H})^+$ and $\varepsilon > 0$. Since a is compact and $\mathcal{H}$ is infinite dimensional, 0 lies in the spectrum of a as an accumulation point or eigenvalue. Hence for every $\delta > 0$ there is a unit vector $\xi \in \mathcal{H}$ with $\langle a\xi, \xi \rangle < \delta$.

Let p_ξ be the rank-one projection onto $\mathbb{C}\xi$. Since the ideal norm is finite on rank-one operators, choose $\delta > 0$ so small that

$$\delta \|p_\xi\|_{\mathcal{I}} < \varepsilon \quad \text{and} \quad \langle a\xi, \xi \rangle < \delta.$$

Then $b = a - \delta p_\xi$ belongs to $\mathcal{I}(\mathcal{H})^{\text{sa}}$ and $\|b - a\|_{\mathcal{I}} < \varepsilon$, but

$$\langle b\xi, \xi \rangle = \langle a\xi, \xi \rangle - \delta < 0.$$

Thus b is not positive. Every ball around a therefore meets the complement of $\mathcal{I}(\mathcal{H})^+$. $\square$

Corollary 1. *The sets $\mathcal{I}(\mathcal{H})_0^+$ and $\{x \in \mathcal{I}(\mathcal{H})^{\text{sa}} : 0 \leq x \leq I\}$ also have empty interior in $\mathcal{I}(\mathcal{H})^{\text{sa}}$.*

Proof. Both sets are contained in $\mathcal{I}(\mathcal{H})^+$, whose interior is empty. $\square$

Theorem 1. *Let $\mathcal{H}$ be infinite dimensional. For the direct countable extensions of the finite examples listed in Proposition 4.17 of the source paper, the following holds.*

- *The full-domain power example $\gamma \sum_i |x_i|^{1/\gamma}$, with $p = 1/\gamma \in (1, \infty)$, gives the positive example $(1/p)\|a\|_{S_p}^p$ on $S_p(\mathcal{H})^{\text{sa}}$.*
- *The following domain-restricted examples cannot satisfy Proposition 4.20's nonempty-interior hypotheses in any infinite-dimensional compact-operator ideal:*
 - *the Kullback–Leibler/Umegaki entropy $\text{Tr}(x \log x - x)$ on the positive compact cone;*
 - *the Burg/Pinsker logarithmic barrier $-\text{Tr}(\log x)$ on the strictly positive compact cone;*
 - *the Fermi–Dirac entropy $\text{Tr}(x \log x + (I - x) \log(I - x))$ on $0 \leq x \leq I$;*
 - *the positive-domain Tsallis/power examples $\text{Tr}(x^\alpha)$ with the positive-cone domain restrictions appearing in the finite list.*

Proof. The positive full-domain statement is the preceding lemma, after the identification $p = 1/\gamma$.

Each listed domain-restricted countable extension has domain contained in one of the empty-interior sets from the lemma and corollary. Consequently

$$\text{int dom}(f \circ \tilde{\lambda}) = \emptyset$$

in the norm topology of $\mathcal{I}(\mathcal{H})^{\text{sa}}$. Proposition 4.20 assumes, among other things, that the relevant interiors of $f \circ \tilde{\lambda}$ and its Fenchel dual are nonempty. Therefore these direct countable extensions cannot satisfy the setup of Proposition 4.20. $\square$

Scope and Limitations

This answers only the direct countable-extension reading of the finite list in Proposition 4.17. It does not rule out:

- other full-domain countable spectral functions beyond the norm-power case isolated above;
- different topologies or non-norm interiors;
- renormalised domains that are not contained in the positive compact cone or compact operator interval;
- other unitarily invariant Euler–Legendre functions with $\Psi \neq \Psi_\varphi$.

Novelty and Literature Check

The cheap run indexes were searched for arXiv:2511.14873, Bregman, Vainberg, relative entropy, Euler–Legendre, and spectral examples, with no duplicate packet found. A web search for the paper title and phrases around “countable extensions” and “Euler–Legendre Schatten” found the source arXiv page but no later explicit answer to Remark 4.21(iii).

Human Review Recommendation

Review whether the source author’s phrase “countable extensions” could mean a non-norm topology or a renormalised domain. Under the norm topology and the natural compact-operator domains displayed in Proposition 4.17, the empty-interior obstruction is decisive.

References

- [1] R.-P. Kostecki, *Vainberg–Bregman relative entropy and quasinonexpansive operators*, arXiv:2511.14873, 2025.